

Fine-tuning Sequence-to-Sequence Language Models for Automated Topic Labeling

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Abstract—Topic modeling is widely employed for knowledge discovery across various domains, yet the output—typically a list of representative keywords—often requires human interpretation through topic labeling. Current approaches rely on either manual annotation, which is time-consuming and subjective, or large language model APIs, which introduce high stochasticity and uncontrolled behavior. We propose fine-tuning specialized sequence-to-sequence models for automated topic labeling. We compare multiple large language models and establish Gemini as a reliable generator of training data. Through a comprehensive expert validation study involving 35 PhD holders across 8 faculties, we achieve an 87.22% agreement rate, statistically validating our data generation methodology. We fine-tune T5-base and T5-large models on two curated datasets: a verified expert-validated dataset (812 pairs) and a taxonomy-based dataset covering 911 research fields (6,377 pairs). Our encoder-freezing strategy preserves semantic representations while enabling task-specific adaptation. Evaluation on a domain-expert test set demonstrates mean cosine similarities ranging from 0.627 to 0.645, with T5-large models achieving superior performance. This work establishes a practical framework for controllable, efficient topic labeling in scientific domains.

Index Terms—Topic modeling, sequence-to-sequence learning, T5, topic labeling, transfer learning, natural language processing

I. INTRODUCTION

Topic modeling has emerged as a fundamental technique for unsupervised knowledge discovery in large text corpora [1]. Methods such as Latent Dirichlet Allocation (LDA) and its variants extract latent thematic structures by identifying co-occurring term patterns. While these approaches effectively discover topics represented as probability distributions over words, they produce outputs that require additional interpretation: a ranked list of topic keywords that stakeholders must manually label to understand their semantic meaning.

This labeling step presents significant challenges in operational deployment. Manual annotation is labor-intensive, requires domain expertise, and introduces subjectivity that hampers reproducibility. Recent practitioners have turned to large language models (LLMs) such as ChatGPT, Claude, and Gemini for automated labeling. However, these general-purpose models exhibit several limitations: high inference costs, unpredictable stochasticity due to their broad knowledge base, lack of fine-grained control, and potential hallucination of labels that may not accurately reflect the topic keywords.

We address these challenges by proposing a specialized approach: fine-tuning compact sequence-to-sequence (seq2seq) models specifically for the topic labeling task. Our method combines the semantic understanding of pretrained transformer models with task-specific adaptation, offering a controllable, efficient, and reproducible alternative to both manual labeling and API-based LLM approaches.

A. Contributions

Our work makes the following contributions:

- 1) **Comparative LLM Analysis:** We systematically evaluate six state-of-the-art LLMs for topic labeling quality using semantic similarity metrics on domain-expert-verified topics.
- 2) **Expert Validation Framework:** We conduct a rigorous validation study with 35 PhD holders across 8 faculties, statistically demonstrating that LLM-generated training pairs achieve 87.22% expert agreement ($p < 0.00001$).
- 3) **Curated Datasets:** We construct two complementary datasets: an expert-verified dataset (812 pairs) and a comprehensive taxonomy-based dataset spanning 911 research fields (6,377 pairs).
- 4) **Specialized Fine-tuning Strategy:** We introduce an encoder-freezing approach for T5 models that preserves semantic representations while enabling efficient task adaptation with limited data.
- 5) **Empirical Evaluation:** We demonstrate that fine-tuned T5 models achieve competitive performance with mean cosine similarities of 0.627–0.645, offering a practical alternative to expensive LLM APIs.

II. RELATED WORK

A. Topic Modeling and Labeling

Classical topic modeling approaches such as LDA [1], Non-negative Matrix Factorization [2], and neural topic models [3] produce distributions over vocabulary terms. Early labeling methods relied on heuristics such as selecting top-ranked terms or extracting noun phrases from topic-relevant documents [4]. These approaches often generated verbose or ambiguous labels.

Recent work has explored neural approaches to topic labeling. Bianchi et al. [5] utilized multilingual sentence embeddings for label generation, while others have leveraged

knowledge graphs [6] or Wikipedia concepts [7] to enhance label quality. However, these methods remain limited in their ability to generate concise, human-readable labels without extensive domain-specific resources.

B. Sequence-to-Sequence Learning

The transformer architecture [8] revolutionized sequence modeling tasks. The Text-to-Text Transfer Transformer (T5) [9] unified various NLP tasks under a single text-to-text framework, demonstrating strong performance across question answering, summarization, and translation. T5’s encoder-decoder structure and extensive pretraining on diverse text corpora make it particularly suitable for conditional text generation tasks.

Transfer learning strategies for T5 have been explored in domain adaptation contexts [10]. Partial parameter freezing has proven effective in preserving pretrained knowledge while enabling task-specific adaptation [11], particularly when training data is limited.

C. LLMs for Text Generation

Recent large language models such as GPT-4 [12], Claude, and Gemini have demonstrated impressive few-shot and zero-shot capabilities across diverse tasks. However, their application to specialized domains faces challenges: high computational costs, difficulty in reproducibility due to model updates, limited control over output characteristics, and potential misalignment with domain-specific requirements. Our work positions fine-tuned compact models as a complementary approach that balances performance with controllability and efficiency.

III. METHODOLOGY

A. LLM Performance Comparison

We initiated our study by comparing the topic labeling capabilities of six prominent LLMs: GPT-4, Claude, Perplexity, Gemini Pro, Grok, and DeepSeek. We utilized a reference dataset of 67 expert-verified topics from our research group’s previous work, where each topic comprised 8 representative keywords and a validated topic label assigned through ChatGPT API and subsequently verified by domain expert supervisors.

For each topic, we generated labels using identical prompts across all six models to ensure fair comparison. To quantitatively assess semantic alignment between generated labels and source keywords, we employed the SPECTER model [13], a Transformer-based scientific document encoder trained on citation graphs. SPECTER was selected for its superior performance on scientific text, aligning with our use case.

We embedded both topic keywords and generated labels using SPECTER, then computed cosine similarity between keyword embeddings and label embeddings for each model. Results demonstrated that Grok achieved the highest mean similarity (0.887), followed closely by Gemini (0.885), Perplexity (0.883), Claude (0.881), GPT-4 (0.878), and DeepSeek (0.873). While Grok marginally outperformed other models,

Gemini Pro followed as a close second with a negligible performance difference. Consequently, we selected Gemini Pro for subsequent data generation due to its seamless integration with Google Sheets through AI-generated columns, facilitating efficient large-scale dataset creation.

B. Expert Validation Study

1) *Hypothesis and Experimental Design:* We formulated the hypothesis: *LLMs can generate high-quality topic keyword-label pairs for specific research fields with expert-level accuracy.* To test this hypothesis rigorously, we designed a comprehensive validation study involving domain experts.

We recruited 35 PhD holders from The German University in Cairo spanning 8 faculties: Media Engineering and Technology, Information Engineering and Technology, Engineering and Materials Science, Pharmacy and Biotechnology, Management Technology, Applied Sciences and Arts, plus Mathematics and Physics departments. For each expert, we identified 3–6 research specializations based on their public research profiles (e.g., Google Scholar, ResearchGate).

2) *Data Generation Protocol:* For each unique research field identified, we employed Gemini to generate 7 topic keyword-label pairs using a carefully designed prompt:

You are an expert research scientist, with a deep focus on [Research Field]. Generate 7 unique (topic_words, topic_label) pairs relevant to this field. Guidelines: (1) topic_words must be a JSON list of 5 lowercase, single-word strings representing highly related keywords as from an LDA topic model. (2) topic_label must be a concise string (2–5 words) accurately summarizing the topic_words. Output as valid JSON.

This prompt structure was designed to elicit focused, domain-relevant pairs while maintaining consistency in format and granularity.

3) *Validation Protocol:* We developed a web-based survey platform where experts evaluated pairs from their assigned research fields. For each pair, participants voted “Agree” or “Disagree” based on whether the keywords and label accurately represented a coherent research topic in their field. To ensure data quality, we embedded trap questions—pairs intentionally designed to elicit disagreement from careful evaluators. Participants who voted “Agree” on all pairs without discrimination were flagged, and their responses were excluded from analysis.

4) *Statistical Analysis:* From 931 evaluated pairs, 812 received “Agree” votes, yielding an agreement rate of 87.22%. We applied a one-sample proportion z-test to determine whether this rate significantly exceeded random chance (50% baseline for binary votes):

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \quad (1)$$

where $\hat{p} = 0.8722$ (observed proportion), $p_0 = 0.50$ (null hypothesis proportion), and $n = 931$ (sample size). This

yielded $Z = 22.71$ with $p < 0.00001$, providing overwhelming statistical evidence that expert agreement significantly exceeds chance. This validates our methodology for using LLMs to generate training data for topic labeling.

C. Dataset Construction

1) *Verified Dataset*: The validated expert survey yielded our first dataset. After filtering pairs that received "Disagree" votes and excluding responses from participants who failed trap questions, we retained 812 high-quality pairs. This **Verified Dataset** represents expert-validated, domain-specific topic labels spanning diverse scientific fields with confirmed accuracy.

2) *Taxonomy Dataset*: To achieve broader coverage across scientific disciplines, we leveraged Elsevier's Digital Commons Three-Tiered Taxonomy of Academic Disciplines. This comprehensive taxonomy organizes knowledge into hierarchical levels: broad disciplines, subdisciplines, and specific research fields.

We targeted the third tier, comprising 913 specific research fields (e.g., Physical Sciences and Mathematics $\rightarrow$ Computer Sciences $\rightarrow$ Software Engineering). Using Gemini with the same prompt structure employed in the validation study, we generated 7 topic keyword-label pairs per field. Two fields ("Holocaust and Genocide Studies" and one other) were excluded as Gemini declined generation due to ethical safeguards, resulting in 911 fields.

This process produced 6,377 topic keyword-label pairs spanning the breadth of academic research, constituting our **Taxonomy Dataset**. While not expert-validated, this dataset provides extensive coverage and diversity, complementing the focused accuracy of the Verified Dataset.

IV. MODEL ARCHITECTURE AND TRAINING

A. Mathematical Formulation

Let a discovered topic be represented by an ordered set of representative keywords:

$$\mathbf{w} = (w_1, w_2, \dots, w_n) \quad (2)$$

where each w_i corresponds to a high-importance term extracted from a topic modeling algorithm. The objective is to generate a concise textual label $\mathbf{y}$ that semantically summarizes the underlying topic:

$$\mathbf{y} = (y_1, y_2, \dots, y_m) \quad (3)$$

1) *Input-Output Mapping*: The task is formulated as a conditional sequence-to-sequence learning problem. A natural language prompt embeds the topic keywords:

$$\mathbf{x} = \text{Prompt}(w_1, \dots, w_n) \quad (4)$$

where $\mathbf{x}$ is the input token sequence. The model learns a conditional distribution:

$$p_\theta(\mathbf{y} | \mathbf{x}) \quad (5)$$

parameterized by θ , the trainable parameters of the T5 model.

2) *Encoder-Decoder Architecture*: The encoder transforms input $\mathbf{x}$ into contextualized hidden states:

$$\mathbf{H} = \text{Encoder}_\theta(\mathbf{x}) = (h_1, h_2, \dots, h_T) \quad (6)$$

where T is the input sequence length and $h_t \in \mathbb{R}^d$ represents a contextual embedding.

The decoder generates output autoregressively. At step t , the probability of token y_t is:

$$p_\theta(y_t | y_{<t}, \mathbf{x}) = \text{Decoder}_\theta(y_{<t}, \mathbf{H}) \quad (7)$$

where $y_{<t} = (y_1, \dots, y_{t-1})$.

The joint probability of the output sequence is:

$$p_\theta(\mathbf{y} | \mathbf{x}) = \prod_{t=1}^m p_\theta(y_t | y_{<t}, \mathbf{x}) \quad (8)$$

3) *Training Objective*: Given dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^N$, the training objective minimizes negative log-likelihood:

$$\mathcal{L}(\theta) = - \sum_{i=1}^N \sum_{t=1}^{m_i} \log p_\theta(y_t^{(i)} | y_{<t}^{(i)}, \mathbf{x}^{(i)}) \quad (9)$$

This corresponds to token-level cross-entropy loss, encouraging accurate and fluent label generation.

B. Fine-tuning Strategy

1) *Model Selection and Initialization*: We adopt the pre-trained T5 architecture due to its unified text-to-text formulation and strong performance on conditional generation tasks. We evaluate both T5-base (220M parameters) and T5-large (770M parameters), initialized from pretrained checkpoints. T5-small was excluded due to insufficient capacity, while larger variants (T5-3B, T5-11B) were deemed impractical given our dataset sizes and risk of overfitting.

2) *Encoder Freezing Strategy*: To preserve pretrained semantic representations while enabling task-specific adaptation, we employ partial encoder freezing. Specifically, the first four encoder layers remain frozen during training. Lower transformer layers typically capture general linguistic and semantic features, whereas higher layers encode task-specific abstractions [11]. Let $\theta = \{\theta_e^{(1:L)}, \theta_d\}$ denote model parameters, where $\theta_e^{(l)}$ represents the l -th encoder layer and θ_d represents decoder parameters. During training:

$$\frac{\partial \mathcal{L}}{\partial \theta_e^{(l)}} = 0 \quad \forall l \in \{1, \dots, 4\} \quad (10)$$

Only upper encoder layers $\theta_e^{(5:L)}$ and decoder θ_d are updated. This constraint prevents catastrophic forgetting of fundamental semantic knowledge while specializing the model for topic labeling.

3) *Optimization Configuration*: We employ the AdamW optimizer with decoupled weight decay. Key hyperparameters:

- Learning rate: 5×10^{-5} (deliberately small for stable fine-tuning)
- Weight decay: 0.01 (regularization)
- Gradient clipping: maximum norm of 1.0
- Warm-up: linear schedule over 10% of training steps

The low learning rate ensures stable fine-tuning and minimizes destructive updates to pretrained weights. Gradient clipping prevents exploding gradients during optimization.

4) *Training Procedure*: Datasets are split into training and validation subsets. Data augmentation is applied exclusively to training data, while validation sets remain unaltered to accurately monitor generalization. Early stopping based on validation loss with patience prevents overfitting. The checkpoint with lowest validation loss is retained as the final model.

5) *Inference and Decoding*: At inference, we employ beam search decoding with constraints to promote diversity and readability:

$$\hat{\mathbf{y}} = \arg \max_{\mathbf{y}} p_{\theta}(\mathbf{y} | \mathbf{x}) \quad (11)$$

Repetition penalties, n-gram blocking, and length constraints prevent trivial copying of input keywords and encourage concise, human-interpretable labels.

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

We fine-tuned four models by combining two architectures (T5-base, T5-large) with two datasets (Verified, Taxonomy):

- **Verified Base**: T5-base on 812 expert-validated pairs
- **Verified Large**: T5-large on 812 expert-validated pairs
- **Taxonomy Base**: T5-base on 6,377 taxonomy pairs
- **Taxonomy Large**: T5-large on 6,377 taxonomy pairs

B. Evaluation Protocol

We constructed a test set of 161 topic keyword sets within the domain expertise of our supervisors, ensuring the test data reflects real-world topic modeling outputs. For each test instance, we generated labels using all four fine-tuned models.

Following our LLM comparison methodology, we embedded both topic keywords and all generated labels using the SPECTER model. Cosine similarity was computed between each set of topic keywords and its corresponding generated label, providing a semantic alignment metric. This evaluation approach measures whether generated labels accurately capture the semantic content of the input keywords rather than requiring exact string matching with reference labels.

C. Results and Analysis

Table I presents comprehensive performance metrics for all four models, and Fig. 1 illustrates the distribution of these similarity scores.

TABLE I
MODEL PERFORMANCE ON TEST SET (COSINE SIMILARITY WITH SPECTER EMBEDDINGS)

Model	Mean	Std	Min	Max
Verified Base	0.628	0.201	0.081	1.000
Verified Large	0.644	0.197	0.085	1.000
Taxonomy Base	0.627	0.206	0.031	1.000
Taxonomy Large	0.645	0.195	0.017	1.000

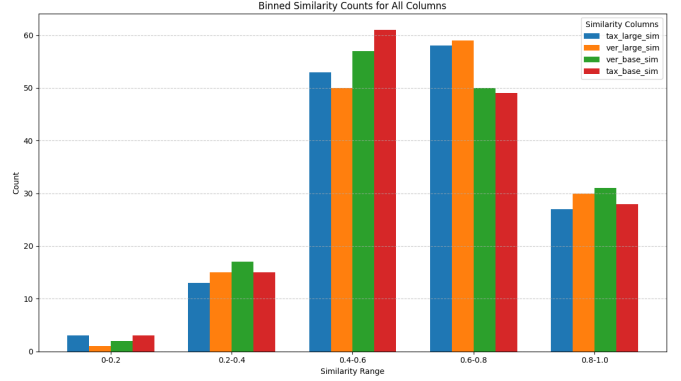


Fig. 1. Binned similarity counts for all models on the test set. The x-axis represents cosine similarity ranges, and the y-axis represents the frequency of topics falling within that range. A shift towards higher similarity bins (0.6–0.8 and 0.8–1.0) indicates better semantic alignment.

1) *Key Observations*: **Model Size Impact**: T5-large models consistently outperformed T5-base counterparts, with Taxonomy Large achieving the highest mean similarity (0.645). The performance gain from base to large models was modest but consistent across both datasets (approximately 0.016–0.018 improvement), suggesting that larger capacity helps capture nuanced semantic relationships.

Distribution Analysis: As shown in Fig. 1, the similarity scores are not uniformly distributed. We observe that the T5-Large models (represented by the Blue and Orange bars) exhibit a higher concentration in the 0.6–0.8 similarity bin compared to the Base models. Conversely, the Taxonomy Base model (Red bar) shows a distinct peak in the lower 0.4–0.6 range. This indicates that the increased parameter count of the Large models does not just improve the average slightly; it actively shifts a significant portion of “acceptable” predictions (0.4–0.6) into the “high quality” range (0.6–0.8), reducing the frequency of mediocre labels.

Dataset Influence: Surprisingly, models trained on the larger Taxonomy Dataset performed comparably to those trained on the smaller but expert-verified dataset. Taxonomy Large marginally outperformed Verified Large (0.645 vs. 0.644), while base models showed nearly identical performance (0.627 vs. 0.628). This suggests that the breadth of coverage in the Taxonomy Dataset compensates for the lack of explicit expert validation, and that Gemini-generated labels are sufficiently reliable for training.

Variance and Stability: All models exhibited similar standard deviations (0.195–0.206), indicating consistent behavior

across diverse topics. The high maximum similarities (1.000 for all models) demonstrate that models can achieve perfect semantic alignment on certain topics, while minimum values reveal edge cases requiring further investigation.

2) *Error Analysis*: Examination of low-similarity cases revealed a systematic failure mode: abbreviations and acronyms. For instance, topic keywords including "SHAP" and "LIME" (standard explainable AI methods) were misinterpreted by models as "shape" and "lime" (the fruit), resulting in semantically incorrect labels. This occurs because:

- 1) T5's subword tokenization treats acronyms as sequences of characters rather than semantic units.
- 2) The pretraining corpus likely contains more instances of common words than technical acronyms.
- 3) Our training datasets, while diverse, may not adequately represent acronym-heavy domains.

This limitation suggests directions for future work, including acronym-aware preprocessing, domain-specific pretraining, or augmentation strategies that explicitly handle technical terminology.

D. Comparison with LLM Baselines

Comparing our fine-tuned models (mean similarities 0.627–0.645) with the LLM baselines from Section III-A (0.873–0.887), we observe a performance gap. However, this comparison must be contextualized:

- LLM evaluations used 67 topics with verified ground-truth labels; our test set comprised 161 diverse topics.
- LLMs require per-query API calls with associated costs and latency; our models enable offline, cost-free inference.
- LLM outputs exhibit stochasticity and require careful prompt engineering; our models provide deterministic, controllable generation.
- Our approach enables customization, privacy, and deployment in restricted environments.

The trade-off between absolute performance and operational advantages positions fine-tuned compact models as a practical alternative for resource-constrained or privacy-sensitive applications.

VI. DISCUSSION

A. Implications for Topic Modeling Pipelines

Our work demonstrates that specialized seq2seq models can be effectively integrated into topic modeling workflows as a labeling component. The approach offers several advantages over existing practices:

Reproducibility: Unlike API-based LLMs subject to updates and version changes, fine-tuned models provide stable, reproducible outputs critical for scientific research and production systems.

Cost Efficiency: After initial training, inference requires no per-query costs, making the approach viable for large-scale applications processing thousands of topics.

Customization: Organizations can fine-tune models on domain-specific data, incorporating proprietary knowledge or terminology absent from general-purpose LLMs.

Privacy: On-premise deployment eliminates concerns about transmitting sensitive topic keywords to external APIs.

B. Limitations and Future Directions

Several limitations warrant acknowledgment and suggest future research directions:

Acronym Handling: As identified in error analysis, technical acronyms remain challenging. Future work should explore acronym-aware tokenization, external knowledge bases (e.g., integrating domain-specific glossaries), or multi-stage approaches that first resolve acronyms before generating labels.

Evaluation Metrics: While cosine similarity with SPECTER embeddings provides a principled semantic measure, it may not fully capture label quality dimensions such as conciseness, distinctiveness, and stakeholder preferences. Incorporating human evaluation and task-specific metrics (e.g., label usefulness in downstream applications) would provide richer assessment.

Cross-Domain Generalization: Our evaluation focused on scientific domains. Assessing performance on news articles, social media, or legal documents would clarify generalization capabilities and identify domain-specific challenges.

Interactive Refinement: Integrating human-in-the-loop mechanisms where domain experts can provide feedback to iteratively improve model outputs could enhance practical utility.

Multilingual Extension: Extending the approach to non-English languages using multilingual T5 variants (mT5) would broaden applicability to global research communities.

C. Broader Impact

Automated topic labeling has implications across multiple domains:

Academic Research: Accelerating literature review and knowledge synthesis by providing interpretable topic summaries from large document collections.

Business Intelligence: Enabling rapid analysis of customer feedback, market trends, and competitive landscapes through interpretable topic extraction.

Digital Humanities: Supporting historians and social scientists in analyzing historical documents and cultural artifacts at scale.

Our approach democratizes access to these capabilities by reducing dependency on expensive proprietary APIs, making sophisticated NLP tools accessible to resource-constrained institutions and individual researchers.

VII. CONCLUSION

We have presented a comprehensive framework for automated topic labeling using fine-tuned sequence-to-sequence models. Our approach addresses critical limitations of existing methods—manual annotation's inefficiency and LLM APIs'

uncontrollability—by specializing compact transformer models for this specific task.

Through rigorous expert validation involving 35 PhD holders, we established that LLM-generated training data achieves 87.22% expert agreement ($p < 0.00001$), validating our data generation methodology. We constructed two complementary datasets: an expert-verified dataset (812 pairs) and a comprehensive taxonomy-based dataset (6,377 pairs spanning 911 research fields).

Our encoder-freezing strategy for T5-base and T5-large models balances semantic preservation with task-specific adaptation. Evaluation on a 161-topic test set demonstrates competitive performance with mean cosine similarities of 0.627–0.645, with T5-large models achieving superior results. While a performance gap exists compared to large-scale LLMs, our approach offers critical operational advantages: reproducibility, cost efficiency, customization, and privacy preservation.

Error analysis revealed challenges with technical acronyms, suggesting concrete directions for improvement. Future work will address these limitations while exploring cross-domain generalization, multilingual extension, and human-in-the-loop refinement mechanisms.

This work establishes fine-tuned seq2seq models as a practical, efficient alternative for topic labeling in diverse scientific and industrial applications, contributing to more interpretable and accessible topic modeling systems.

ACKNOWLEDGMENT

The authors thank the 35 PhD holders from The German University in Cairo who participated in the expert validation study.

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